96.33% accuracy for CIFAR-10 and 79.82% accuracy for ImageNet. We find similar observations on other diversity metrics, including CK and KW. This motivates us to investigate how to properly measure ensemble diversity and optimize the ensemble pruning process.

Table 6. Baseline Ensemble Pruning by Mean Threshold (CIFAR-10)

Methods	BD>0.339 (baseline)	F-BD>0.602 (optimized)	GD>0.524 (baseline)	F-GD>0.543 (optimized)
Acc Range (%) of *GEnsSet*	93.56∼96.72	**95.71**∼97.15	93.56∼97.15	**95.71**∼97.15
Precision (%)	6.18	**63.53**	36.90	**53.90**
Recall (%)	11.72	**95.51**	63.10	**97.59**
Cost Reduction	10%∼80%	10%∼80%	10%∼80%	10%∼80%

Table 7. Baseline Ensemble Pruning by Mean Threshold (ImageNet)

Methods	BD>0.314 (baseline)	F-BD>0.632 (optimized)	GD>0.335 (baseline)	F-GD>0.594 (optimized)
Acc Range (%) of *GEnsSet*	61.39∼80.54	**76.77**∼80.77	70.79∼80.60	**76.41**∼80.77
Precision (%)	8.53	**41.93**	17.74	**36.75**
Recall (%)	23.15	**98.52**	46.31	**99.01**
Cost Reduction	10%∼80%	10%∼80%	10%∼80%	10%∼80%

Focal Diversity based Hierarchical Pruning. We improve the ensemble diversity measurement and ensemble pruning process with three novel techniques in this paper. *First,* we use focal diversity metrics to optimize the ensemble diversity measurements, which utilizes the concept of focal models to sample negative samples and precisely capture the failure independence of member networks of a deep ensemble team. *Second,* we introduce a novel hierarchical pruning approach, which leverages focal diversity metrics and iteratively prunes out subsets of redundant member networks with low diversity from the entire ensemble. *Third,* we combine multiple focal diversity metrics through focal diversity consensus voting to further enhance the hierarchical ensemble pruning performance.

3 FOCAL DIVERSITY BASED HIERARCHICAL PRUNING

3.1 Focal Diversity Concept and Algorithm

Our focal diversity metrics are designed to provide more accurate measurements of failure independence of the member networks in a deep ensemble team, including both pairwise (F-CK and F-BD) and non-pairwise (F-GD and F-KW) diversity metrics. Unlike traditional ensemble diversity evaluation approaches, which randomly select samples from the validation set to evaluate ensemble diversity, our focal diversity measurements draw random negative samples from $NegSampSet(F_f)$ based on a specific focal model F_f on which the focal model F_f makes prediction errors and then calculate the focal negative correlation score. For a given deep ensemble of S member networks, each of the S member networks will serve as the focal model (F_f) to draw negative samples. Therefore, we have a total of S focal negative correlation scores, where each corresponds to a member network serving as the focal model. Then we obtain the focal diversity score for this deep ensemble of S member networks through the (weighted) average of S focal negative correlation scores. The focal model concept is motivated by ensemble defense against adversarial attacks [9, 34], where the attack victim model is protected in a defense ensemble team. The focal model can be viewed as the victim model to evaluate the failure independence of other member models to this focal model in an ensemble team, i.e., the focal negative correlation score. Thus, the ensemble diversity score for

this ensemble of size S can be computed as the (weighted) average of S focal negative correlation scores by taking each member model as a focal model to mitigate the bias in diversity evaluation using only one focal model. Our preliminary results have demonstrated promising performance of the focal diversity metrics in capturing the failure independence of the member networks of a deep ensemble team [37–39].

Algorithm 1 Focal Diversity Metric Calculation

1: **procedure** GETFQ($NegSampSet, Q, EnsSet$)
2: **Input**: $NegSampSet$: negative sample sets for each focal model F_f; Q: the focal diversity metric, such as F-CK, F-BD, F-KW and F-GD; $EnsSet$: the set of candidate sub-ensemble teams to be considered;
3: **Output**: FQ: focal diversity scores
4: Initialize $D(Q) = \{\}$, $\overline{D}(Q) = \{\}$
5: Initialize $FQ = \{\}$ ▹ A map of diversity scores and teams
6: **for** $S = 2$ to $M - 1$ **do**
7: **for** $f = 0$ to $M - 1$ **do**
8: Obtain $EnsSet(F_f, S)$ with candidate sub-ensembles of size S and containing the focal model F_f.
9: Initialize $D(Q, S, F_f) = \{\}$
10: **for** $i = 1$ to $|EnsSet(F_f, S)|$ **do**
11: ▹ calculate the focal negative correlation score for $T_i \in EnsSet(F_f, S)$ using the definition of Q
12: $q_i = FocalNegativeCorrelation(Q, T_i, NegSampSet(F_f))$
13: $D(Q, S, F_f)$.append(q_i)
14: **end for**
15: **for** $i = 1$ to $|EnsSet(F_f, S)|$ **do**
16: ▹ scale focal negative correlation scores for sub-ensembles of the same size S
17: $\overline{D}(Q, S, F_f, T_i) = \frac{q_i - \min(D(Q,S,F_f))}{\max(D(Q,S,F_f)) - \min(D(Q,S,F_f))}$
18: **end for**
19: **end for**
20: Obtain $EnsSet(S)$ with candidate sub-ensemble teams of size S
21: **for** $i = 1$ to $|EnsSet(S)|$ **do**
22: Initialize $tmpD = \{\}$
23: **for** $j = 1$ to $|T_i|$ **do**
24: $tmpD$.append($\overline{D}(Q, S, F_f = T_i[j], T_i)$)
25: **end for**
26: ▹ Obtain the member model accuracy ranks as the member model weights.
27: $w = MemberModelAccuracyRank(T_i)$
28: $FQ(T_i) = WeightedAverage(w, tmpD)$
29: **end for**
30: **end for**
31: **return** FQ
32: **end procedure**

In general, our focal diversity metrics compare the ensemble diversity scores among the ensembles of the same size S, randomly draw negative samples from each focal model (F_f, i.e., $NegSampSet(F_f)$) and calculate the focal diversity score as the average of S focal negative correlation scores by taking each member model as the focal model. Algorithm 1 gives a skeleton of calculating the focal diversity scores for all the candidate sub-ensembles in $EnsSet$. For each team size S (Line 6~30), we follow two general steps to calculate the focal diversity scores for each ensemble. *First,* for each member model F_f, let $EnsSet(F_f, S)$ denote all candidate ensembles

of size S, each containing the focal model F_f. We first compute the focal negative correlation score for each ensemble in $EnsSet(F_f, S)$ with the negative samples randomly drawn from the focal model F_f ($NegSampSet(F_f)$) and store them in $D(Q, S, F_f)$ (Line 10∼14). Then, in order to make them comparable across different member models (F_f), we scale $D(Q, S, F_f)$ into $[0, 1]$ and store them in $\overline{D}(Q, S, F_f, T_i)$ for each ensemble team T_i (Line 15∼18). *Second,* for each candidate sub-ensemble (T_i) of size S, we perform a weighted average of the scaled focal negative correlation scores $\overline{D}(Q, S, F_f = T_i[j], T_i)$ associated with each of its focal (member) model $F_f = T_i[j]$ to obtain the focal diversity score. The weight is calculated with the corresponding rank of the accuracy of the member model ($T_i[j]$) in the ensemble team (T_i), i.e., the member models with higher accuracy will have higher weights (Line 21∼29). In addition, we subtract the CK value from 1 when using the CK formula [25] to present the consistent view that high diversity values correspond to high ensemble diversity.

We show a visual comparison between our focal diversity metric F-GD in Figure 1b and the baseline GD metric in Figure 1a for CIFAR-10. Here the baseline mean threshold based ensemble pruning method is used for both F-GD and GD diversity metrics. Compared to the GD based ensemble pruning, the F-GD powered pruning obtains better ensemble pruning results and identifies a larger portion of sub-ensembles (on the right side of the vertical red dashed line) with ensemble accuracy above the reference accuracy of 96.33% (horizontal black dashed line).

When the focal diversity metrics are used in our hierarchical pruning algorithm with a desired team size S_d, we only calculate the diversity scores for the ensembles of size $2 \sim S_d$ by replacing $M - 1$ with S_d in Line 6 of Algorithm 1 to reduce the computation cost. Compared to the baseline diversity metrics, our focal diversity score computation takes about 1.2∼2.4× the time of the baseline diversity score computation for all candidate sub-ensembles in $EnsSet$. The computation of all focal diversity metrics can be finished in several seconds on MNIST, CIFAR-10 and Cora and in several minutes on ImageNet (see Table 16). Given that our focal diversity metrics significantly outperform the baseline diversity metrics in capturing the failure independence of a group of member models, it is beneficial and worthwhile to use our focal diversity metrics to measure the ensemble diversity and identify high quality sub-ensembles. In addition, our hierarchical pruning can effectively prune out about 80% of the ensemble teams in $EnsSet$, which also compensates for the increased computation time of the focal diversity metrics, making our entire solution very efficient.

We show the baseline mean threshold based ensemble pruning results using our focal diversity F-BD and F-GD in the 3rd and 5th columns of Table 6 for CIFAR-10 and Table 7 for ImageNet. Both F-BD and F-GD (optimized) substantially outperform the baseline BD and GD based ensemble pruning, as measured by the accuracy range of the selected sub-ensembles in $GEnsSet$, precision and recall. For the other two diversity metrics, CK and KW, we observed similar performance improvements of using our focal diversity metrics, F-CK and F-KW, in the baseline ensemble pruning. Even though our focal diversity metrics can significantly improve the baseline mean threshold based ensemble pruning method, achieving very high recall of over 95%, we can still enhance the 63.53% precision of F-BD and 53.90% precision of F-GD for CIFAR-10 and the 41.93% precision of F-BD and 36.75% precision of F-GD for ImageNet to more accurately identify high-quality sub-ensembles. In addition, the baseline mean threshold based ensemble pruning examines every sub-ensemble in $EnsSet$ with high computational cost. These observations have motivated us to explore new methods to enhance ensemble pruning efficiency and accuracy.